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Uncovering patterns of converging human-induced pressure on global lands

Bryam Orihuela-Pinto^{1,5,*} , Patrick W Keys^{1,2} , Frances V Davenport³ and Elizabeth A Barnes^{1,2,4}

¹ Department of Atmospheric Science, Colorado State University, Fort Collins, CO, United States of America

² Department of Earth and Environment, Boston University, Boston, MA, United States of America

³ Department of Civil and Environmental Engineering, Colorado State University, Fort Collins, CO, United States of America

⁴ Computing and Data Sciences, Boston University, Boston, MA, United States of America

⁵ Current address: Amazon Sustainability Research Institute, Universidad de Ingeniería y Tecnología, Lima, Peru.

* Author to whom any correspondence should be addressed.

E-mail: lorihuela@utec.edu.pe

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Abstract

Monitoring how human activities and other environmental pressures jointly affect different parts of the Earth is critical for guiding conservation, sustainable development, and climate adaptation efforts. Yet, high-resolution, temporally consistent tools to track these combined pressures globally remain limited. Using the newly developed machine learning Human Footprint Index version 2 (ml-HFIV2.0), we conduct an analysis of the intersection between human pressure on land and climate change. The ml-HFIV2.0 was developed using a convolutional neural network trained on Landsat satellite data, building upon a previous version. This new version features a finer spatial resolution of 300 meters and the ability to maintain temporal continuity using Landsat imagery as input. These methodological advancements result in a fully self-sufficient workflow, enabling retrospective and up-to-date assessments of human pressure on global land. Moreover, ml-HFIV2.0 effectively captures rapidly evolving land-use changes, such as illegal mining activities, that were previously overlooked by conventional datasets. Our analysis shows that natural areas, which cover most of the world, exhibit disproportionately higher warming, with 58.5% of these regions warming faster than the global median. In contrast, land areas experiencing some degree of human pressure have typically experienced slower warming, with around 65% of these areas warming at a rate below the global median. Approximately 2.7% of global land area has experienced both high human pressure and rates of warming above the global median. Our findings underscore the importance of analyzing human and climatic pressures simultaneously, highlighting how different land-use categories coincide with global warming patterns. The ability of ml-HFIV2.0 to provide high resolution, up-to-date monitoring offers critical advantages for understanding and managing anthropogenic impacts and their interactions with climate change. These insights are essential for informing conservation, urban planning, and climate adaptation strategies.

1. Introduction

Anthropogenic activities are reshaping Earth's landscapes and ecosystems at an unprecedented scale (Western 2001). Urbanization, agriculture, resource extraction, and energy production have intensified human pressure on natural environments, leading to cascading consequences such as biodiversity loss, habitat fragmentation, and increased risks to human societies (Zipperer *et al* 2020). These transformations are neither uniform nor evenly distributed across the globe. Regions differ in the rate and intensity of changes, shaped by variations in land use, climate, and socio-economic development. For instance, some areas could exhibit rapid ecosystem degradation and warming, while others could show

slow warming despite escalating human pressures over land. Understanding and capturing this heterogeneity is critical for designing effective, region-specific strategies.

Quantifying human influence on land has been approached through various methodologies. Studies that focus on a single pressure, such as nighttime lights (e.g. Mu *et al* 2021) or population density (e.g. Wang *et al* 2018) are limited in capturing the combined effects of multiple human activities. While these datasets provide valuable tools for quantifying certain aspects, their ability to comprehensively represent human pressure varies. For example, nighttime light data offer rapid insights into urban expansion but fail to account for human activities in agricultural, forestry, or remote regions. Similarly, population density maps highlight human presence but do not directly reflect infrastructure development or land-use changes. A more holistic and temporally consistent approach is required to integrate these diverse pressures and provide a more complete representation of human influence on the landscape.

Indices such as the Human Footprint Index (HFI; e.g. Venter *et al* 2016a, Sanderson *et al* 2002, 2022, Williams *et al* 2020) provide an integrated measure of human pressure on the global land surface. These previous efforts to create an HFI rely on datasets from remote sensing and ground-survey data to get information about the extent of built environments, population density, electric infrastructure, crop lands, pasture lands, roads, railways and navigable waterways to generate a comprehensive assessment of human influence. The HFI has proven invaluable in monitoring biodiversity (Venter *et al* 2016b), assessing habitat fragmentation (Watson *et al* 2016), and policymaking (Watson and Venter 2019). However, because these datasets are typically updated at different temporal intervals, traditional HFI approaches may face challenges related to temporal consistency and the timeliness of information. As a result, monitoring long-term trends and detecting emerging environmental changes in near real-time remain ongoing challenges.

In addition to human land pressures, climate change's associated hazards (e.g. heatwaves, droughts, and floods) are intensifying, profoundly impacting both ecosystems and human communities (IPCC 2023). These hazards are inextricably linked to human activities, yet there has been comparatively little work explicitly linking human pressure indices and climate hazards. Recent studies (e.g. Hoffmann *et al* 2019, Khelifa *et al* 2022, Sengupta *et al* 2024) have begun to explore these connections, but significant gaps remain in understanding how land-based pressures and climate trends intersect.

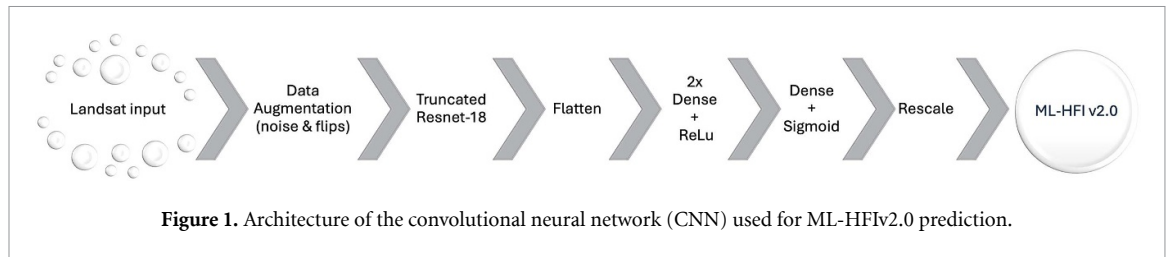
This study provides new insight into co-occurring land surface and climate changes through the use of a novel machine learning-based HFI (ml-HFI). The ml-HFI is based on the approach of Keys *et al* (2021) using a convolutional neural network (CNN) trained on an existing HFI dataset, with Landsat imagery as input features. This new version, referred to as ml-HFIv2.0, features a 300 m spatial resolution (an improvement over the 1200 m resolution of the first version) and offers enhanced control over the data, incorporating several improvements that ensure temporal continuity and enable the use of Landsat data to extend analyses both as far back in time as Landsat data is available and to generate the most up-to-date global HFI maps. Leveraging the ml-HFIv2.0 alongside long-term temperature trends, we provide an integrated framework to assess how human and climate pressures intersect. This integration offers a more comprehensive understanding of anthropogenic impacts on the environment and highlights critical regions where climate and human pressure on land overlap. While both are influenced by human activities, this study does not attempt to establish causal relationships between them, but instead focuses on identifying spatial patterns of co-occurrence. Areas identified to have a high co-occurrence of both climate and land use pressure could be the focus of future studies aimed at informing land management and climate mitigation strategies.

2. Methods

2.1. Landsat data

Landsat imagery serves as the input features for our CNN model. The acquisition and processing of Landsat images for this study were conducted using Google Earth Engine (GEE), which provides access to various Landsat surface reflectance collections. Multiple built-in GEE functions were utilized for cloud, snow, and water masking, as well as for image scaling and compositing.

We obtain the Landsat data from the collection 2, Tier 1, surface reflectance (L2) on the GEE platform. First, we masked cloud, water, and snow pixels using the 'QA_PIXEL' band, which applies predefined bitmask values to exclude undesirable pixels. To generate composite images, cloud-free and snow-free pixels for a given year and region were loaded. When multiple collections were available, they were merged to enhance data coverage. To further refine data quality, we calculate the normalized difference vegetation index and use it to select the greenest 30% of days for each pixel, from which the



median value was extracted. Additionally, imagery from multiple years was integrated, prioritizing cloud-free and snow-free pixels from the current year. If such pixels were unavailable, pixels from the previous year were utilized to ensure comprehensive coverage. Finally, water pixels were identified, and for model training purposes, a single water mask was applied with a buffer area of approximately 4 km away from land of approximately 4 km surrounding the water-land boundaries. This buffer aimed to account for potential artificially created land or changes in water extent over time. This multi-step processing approach resulted in high quality, cloud-free and snow-free composite images for each year from 1999 to 2024 leveraging the consistent and complete Landsat data available from Landsat 7 and subsequent missions.

2.2. ML-HFIv2.0 model development and training

The ml-HFIv2.0 ranges from 0 to 100, where 0 represents intact natural areas and higher values indicate increasing human pressure. To derive this index, we build upon the approach by Keys *et al* (2021) while eliminating previous dependencies on externally processed datasets, resulting in a fully self-sufficient workflow that only requires processed Landsat data for predicting the ml-HFIv2.0. Our model employs a CNN initialized with a ResNet18 architecture (He *et al* 2016), pretrained on ImageNet. The model applies a custom rescaling layer before feature extraction. We retain all layers up to the third-to-last layer of the ResNet18 backbone, keeping them fully trainable. The extracted features are flattened and passed through two dense layers (each with 128 nodes and ReLU activation functions), followed by a final dense layer with a Sigmoid activation to produce the output. Input noise and data augmentation (random flips) are applied during training to improve generalization. This architecture (figure 1) enables the model to leverage spatial context from satellite imagery while maintaining flexibility for pixel-level regression.

The labels are the HFI developed by Sanderson *et al* (2022) which they call Human Impact Index, and whose second-generation version provides annual 300 m resolution labels from 2015 to 2020. Note that ml-HFIv2.0 primarily captures persistent, human activities that are visible through changes in surface reflectance (e.g. urban growth, roads, deforestation), while indirect pressures like population density or energy use are inferred through their association with land surface changes. Therefore, activities like short-term population fluctuations or temporary reductions in human pressure (such as those during COVID-19 lockdowns) that do not leave lasting physical changes on the land are less likely to be detected.

The input to the CNN consists of a $110 \times 110 \times 3$ array of 30 m Landsat pixels, where the third dimension represents the three Landsat spectral bands chosen (i.e. the green, near infrared and short-wave infrared bands). The output of the CNN is the ml-HFIv2.0 value for a single 300 m pixel at the center of the input window. The CNN outputs a single value between 0 and 1, which is later scaled to 0–100. The model is trained to minimize the square root of the mean squared error between the predicted ml-HFIv2.0 and the Sanderson-HFI during the training period. We also apply extra weight to the loss for values near zero, given the importance of predicting zeros accurately. This adjustment was implemented through a conditional weighting scheme, where targets below or equal to 5 were scaled by a factor of 5, while other values retained the default weight of 1.

For model training, we implement a series of strategies. We used stochastic gradient descent as the optimizer and early stopping if the validation loss does not improve for 5 consecutive epochs. To choose our training samples, we use an equal sampling strategy by randomly selecting approximately 1% of available pixels from 2015 to 2019, ensuring an equal number of samples within 10-unit bins of HFI values (e.g. 0–10, 10–20, ..., up to 100). The 1% totals 12 843 627 samples. Of these, 12 330 976 were training samples and 512 651 were validation samples. The validation samples were only used for early stopping during training. For testing and robust evaluation of the model performance we used an entire year of independent data (for the year 2020). None of this data was used for training or early stopping. The years from 2015 to 2019 were chosen to align with the continuous, annually updated HFI data

provided by Sanderson *et al* (2022). This approach enables the ml-HFIv2.0 to be computed for any year with Landsat data. We then evaluate the ml-HFIv2.0 using Landsat data from 1999 to 2024 to capture over two decades of human pressure dynamics.

2.3. Analysis of combined human footprint and climate pressure

To assess the intersection of human pressure and climate change, we analyze 2 m air temperature trends using ERA5-Land data (Muñoz-Sabater *et al* 2021) at a 0.1° spatial resolution, covering the period from 1971 to 2023. Temperature trends were calculated by first computing annual means from monthly data for each grid point, followed by applying linear regression to estimate the slope, which effectively captures long-term trends while minimizing the influence of seasonal variability. This extended timespan starting in 1970 captures the fastest warming 50 yr period in the past 2000 yr (IPCC 2023), providing a robust foundation for evaluating long-term climate trends.

To facilitate a more objective analysis using the dimensionless ml-HFIv2.0 index, we classified land into three categories: ml-HFI values of 0–10 were assigned to the category of ‘natural areas’, ml-HFI values from 10–35 were assigned to the category of ‘used lands’, and ml-HFI values ≥ 35 are classified as ‘settlements’. Natural areas include locations with little to no human influence such as intact forest or desert. Used lands include areas with moderate human influence such as croplands, rangelands, and peri-urban zones. Settlements include areas with high human influence such as densely developed urban environments. These three categories allow us to broadly group locations into regions of low, medium, and high human pressure. Similar yet more complex classifications, such as the Anthrome framework (Ellis *et al* 2020) exist. Using the Anthrome framework, Sanderson *et al* (2022) found that urban areas having a mean HFI value of 40, wild areas a mean HFI value of 1, and multiple other classes exhibiting intermediate values.

The ml-HFIv2.0 values also align broadly with the values found by Sanderson *et al* (2022), though some differences exist due to the nature of the ml-HFIv2.0. Although trained on the Sanderson HFI, the ml-HFIv2.0 produces different values in certain cases, as it is derived from satellite imagery and may detect differently what is captured in the original dataset. We manually verified that the chosen thresholds largely correspond to the intended three land-use categories.

To integrate the datasets and find combined pressures, we first reprojected the ERA5-Land temperature trends to match the coordinate reference system of the ml-HFIv2.0 data. Rather than resampling the temperature trends to the higher ml-HFIv2.0 resolution, we performed a spatial overlay: for each 300 m ml-HFIv2.0 pixel, we extracted the temperature trend value from the corresponding ERA5-Land grid cell. We then classified each pixel based on its ml-HFI value and the overlapping temperature trend. This approach preserves the native resolution of both datasets and enables meaningful aggregation without artificial upscaling or downscaling.

By integrating ml-HFIv2.0 data with temperature trends, this analysis uncovers how spatial patterns of land-use pressures and climate impacts intersect, offering a comprehensive perspective on the compounded effects of anthropogenic activities on environmental and climate systems.

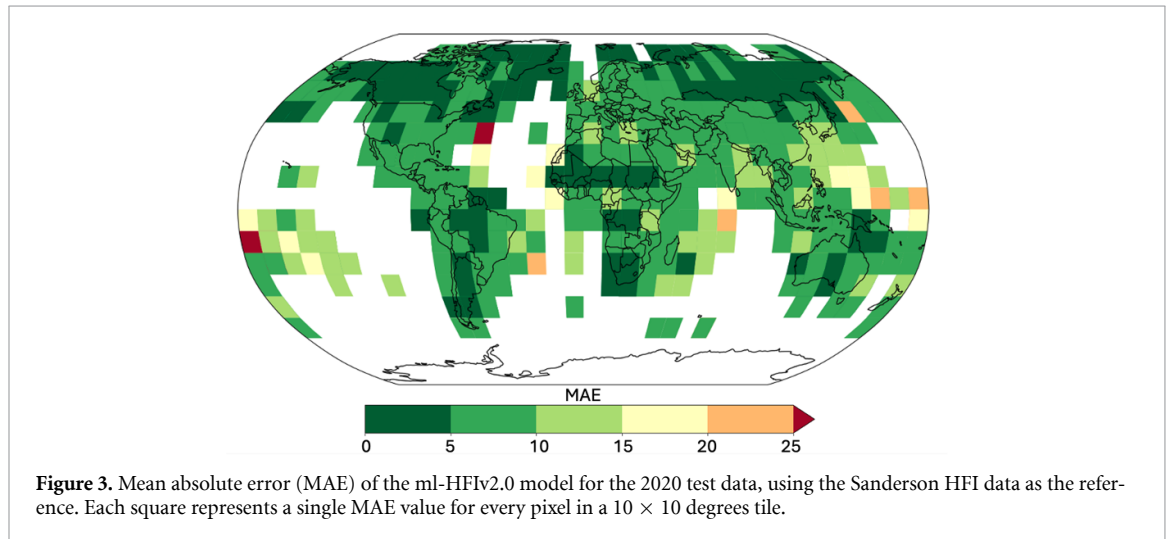
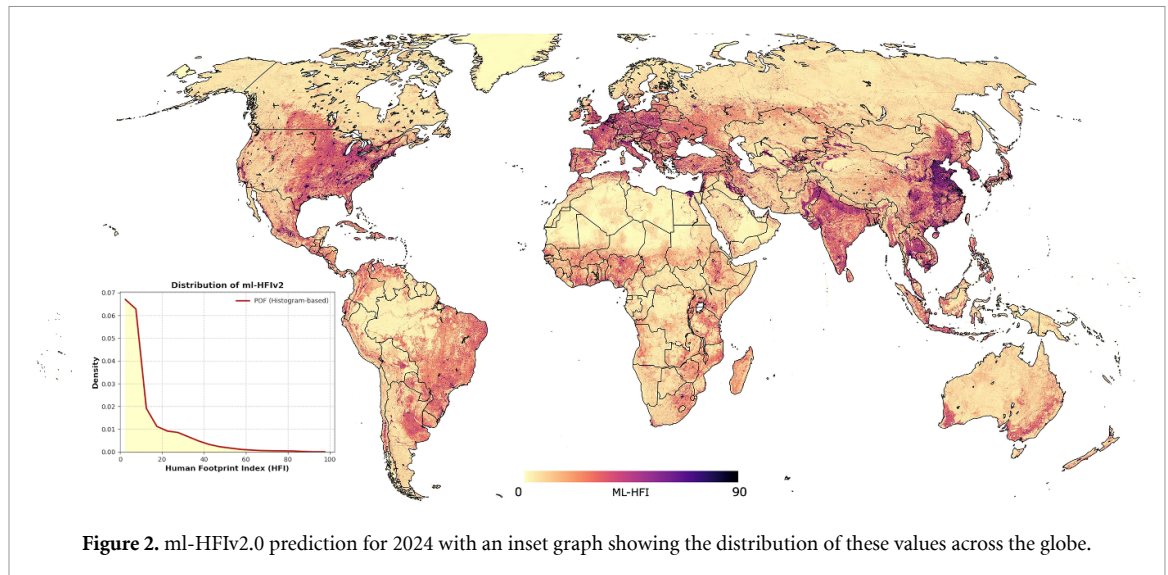
3. Results

3.1. ml-HFIv2.0 and its performance

Figure 2 illustrates the ml-HFIv2.0 for 2024, offering a detailed and up-to-date view of global human pressure on terrestrial environments. This map provides an essential snapshot of the current degree of human influence activities on land, identifying regions with varying levels of impact. The inset graph illustrates the predominance of areas with low values, while regions with high values are relatively rare.

The ml-HFIv2.0 reveals a broad gradient of human influence globally, with densely developed regions exhibiting higher values, while more remote or less industrialized areas displaying lower values. At a broad scale, elevated ml-HFIv2.0 values are observed in highly developed regions such as Europe, the United States, and China, which have experienced significant urban expansion since the late 20th and early 21st centuries (Behnisch *et al* 2022). Additionally, high values are evident closer to the coasts in regions such as Africa and South America, reflecting the rapid increase in coastal urbanization (Cosby *et al* 2024). While some areas remain relatively undisturbed, localized pressures such as deforestation, mining, agricultural expansion, among others have altered natural landscapes. The high-resolution nature of the ml-HFIv2.0 allows for a more detailed examination of these patterns, providing insights beyond broad-scale trends (see section 3.2).

To assess the similarity between the ml-HFIv2.0 and the Sanderson HFI, we computed the mean absolute error (MAE) for 2020, a year excluded from the training data. Figure 3 presents the MAE values across $10^\circ \times 10^\circ$ tiles, showing that most tiles exhibit errors below 10. Higher errors are observed

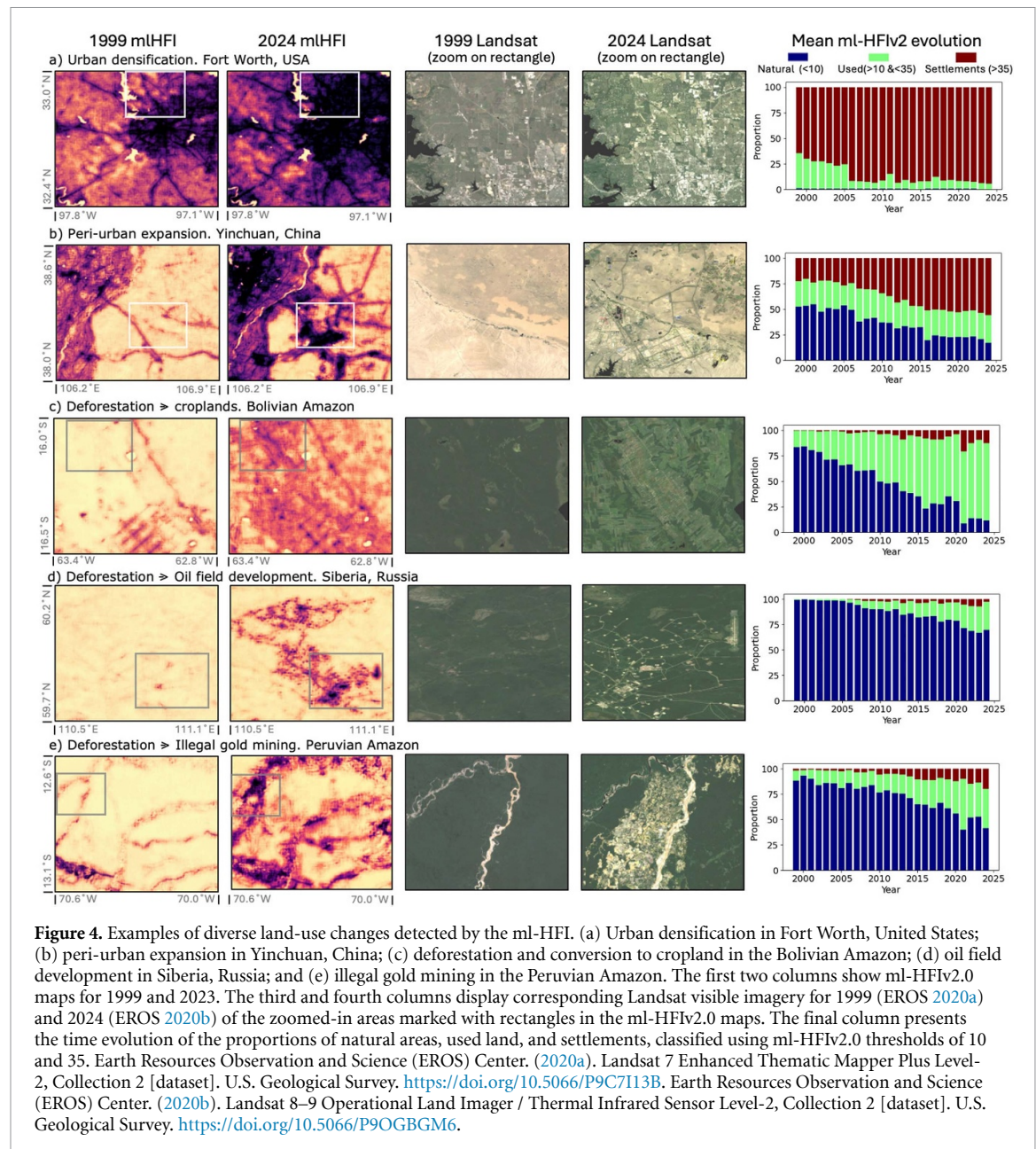


primarily in tiles that include both land and ocean, containing a reduced number of land pixels compared to fully inland tiles contributing to increased error variability. A couple of ocean-containing tiles show very high mean errors (red squares in figure 3), due to the disproportionately high errors associated with the limited number of land pixels within those tiles. Differences between the ml-HFIv2.0 and original HFI dataset arise for multiple reasons. Because the ml-HFIv2.0 is based purely on Landsat imagery and does not rely on other datasets, it can capture features not represented in the Sanderson HFI, such as illegal mining (figure S1(a)). Additionally, the ml-HFIv2.0 does not uniformly emphasize certain features, such as roads, which are systematically detected in the Sanderson HFI due to the use of OpenStreetMap data (Sanderson *et al* 2022). In contrast, our approach identifies roads based on their visible characteristics in Landsat imagery. As a result, the contribution of roads to human pressure varies depending on their context; for example, distinguishing a rural dirt road from a major urban highway. Consequently, we find that some roads, particularly those in remote or sparsely populated areas, are assigned lower values in the ml-HFIv2.0 than in the original Sanderson HFI (figures S1(b) and (c)). More specific examples of ml-HFIv2.0 predictions are provided in the next section.

3.2. ML-HFIv2.0 capabilities

Figure 4 showcases the precision and versatility of the ml-HFIv2.0 dataset through selected examples from the United States, China, Bolivia, Russia, and Peru. These case studies highlight the dataset's capacity to capture diverse human pressures and land-use changes over time, facilitated by its high spatial resolution and temporal continuity.

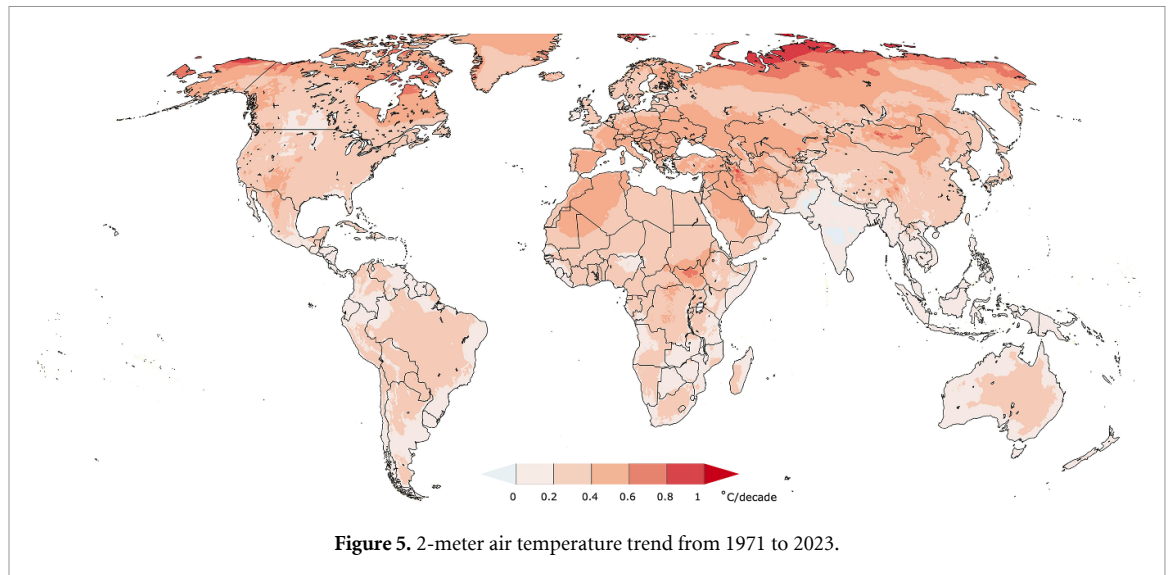
In the United States, the case of Fort Worth, Texas (figure 4(a)) exemplifies increasing urban density during the evaluation period. Most (80%) of the U.S. population were living in urban areas by 2000



(U.S. Census Bureau 2012), and Fort Worth reflects ongoing growth. Notably, ml-HFIv2.0 values peak in 2006, coinciding with a peak in residential construction spending (U.S. Census Bureau n.d.). While a detailed analysis would be needed to confirm a direct causal link, this temporal alignment suggests the model may be capturing signals of development trends. A similar peak is observed when analyzing the temporal evolution of ml-HFIv2.0 in other parts of the United States (not shown).

The city of Yinchuan, China (figure 4(b)), illustrates an example of peri-urban expansion (i.e. urban areas encroaching into non-urban areas). China experienced substantial increases in ml-HFIv2.0 throughout the study period, often due to urban expansion into peripheral rural areas (Chen *et al* 2024). In Yinchuan there is also urban densification, but in the selected area (Landsat images in figure 4(b)) in the outskirts of the city there was a transition from desert terrain to a landscape dominated by industrial facilities, agricultural fields, and new residential neighborhoods. This transformation is corroborated by a steady increase in ml-HFIv2.0 values over time, reflecting rapid and consistent land-use changes from 1999 to 2024.

In the Bolivian Amazon, the department of Santa Cruz has been a deforestation hotspot since the late 1980s. The ml-HFIv2.0 effectively captures this ongoing forest-to-agriculture conversion (figure 4(c)), driven by large-scale soya cultivation as well as other crops such as rice, corn, wheat, sugarcane, and cotton (De La Vega-Leinert and Huber 2019). This transition is reflected in the ml-HFIv2.0



values and corroborated by the time evolution graph, which depicts a decline in natural areas alongside an increase in used lands.

In Siberia, Russia, the development of an oil field and an airport within previously natural areas is clearly detected by the ml-HFIV2.0 (figure 4(d)). The dataset accurately identifies these industrial expansions, with higher ml-HFIV2.0 values corresponding to the central facilities and airport infrastructure that can be seen in the Landsat image inset.

A distinct advantage of the ml-HFIV2.0 is its capacity to detect changes that may be missed by traditional human footprint indices. This is evident, for example in the Peruvian Amazon (figure 4(e)), where the ml-HFIV2.0 exposes illegal gold mining activities (e.g. Cortés-McPherson 2019, Finer *et al* 2023), underscoring its potential to reveal unreported or rapidly evolving land-use changes that are often absent in conventional datasets.

Collectively, these case studies emphasize the ml-HFI's high temporal and spatial resolution, enabling continuous monitoring of human-induced environmental transformations. The dataset's ability to detect both large-scale and subtle changes enhances its utility for supporting sustainable development and land management strategies.

3.3. Temperature trends 1971-2023

Complementing the ml-HFIV2.0 analysis, we examine trends in temperature over the past half century. The 53 yr temperature trend from 1971 to 2023 serves as an indicator of long-term climatic changes which have anthropogenic influence. This approach is supported by evidence that global surface temperatures have risen more rapidly since 1970 than during any other 50 yr period in at least the past 2000 yr (IPCC 2023).

Figure 5 displays the decadal trends in 2 m air temperature over land from 1971 to 2023, illustrating significant variations in warming across different regions. The map highlights areas with accelerated temperature increases, with the lowest values in the southern hemisphere and highest values in the Northern Hemisphere particularly near the Arctic where Arctic amplification accelerates warming (e.g. Serreze and Francis 2006, England *et al* 2021, Estrada *et al* 2021). Overall, the broad warming pattern aligns with previous studies (Hansen *et al* 2010) with certain regions exhibiting distinct warming patterns. For example, India shows slower warming rates partly due to high aerosol concentrations that reflect sunlight and mitigate temperature increases (e.g. Basha *et al* 2017). Conversely, arid and semi-arid regions, such as Northern Africa and the Middle East, experience accelerated warming, a phenomenon known as desert amplification (e.g. Zhou 2016).

The results are also consistent with the broader scientific consensus that human activities have unequivocally warmed the atmosphere, ocean, and land, contributing to changes in climate extremes worldwide (IPCC 2021). The increasing evidence linking human influence to extreme events such as heatwaves, heavy precipitation, droughts, and tropical cyclones has been further strengthened over time (IPCC 2021). Given this well-established connection, analyzing warming patterns could provide insights into the spatial distribution and intensity of climate hazards, which have direct implications for environmental and socio-economic systems.

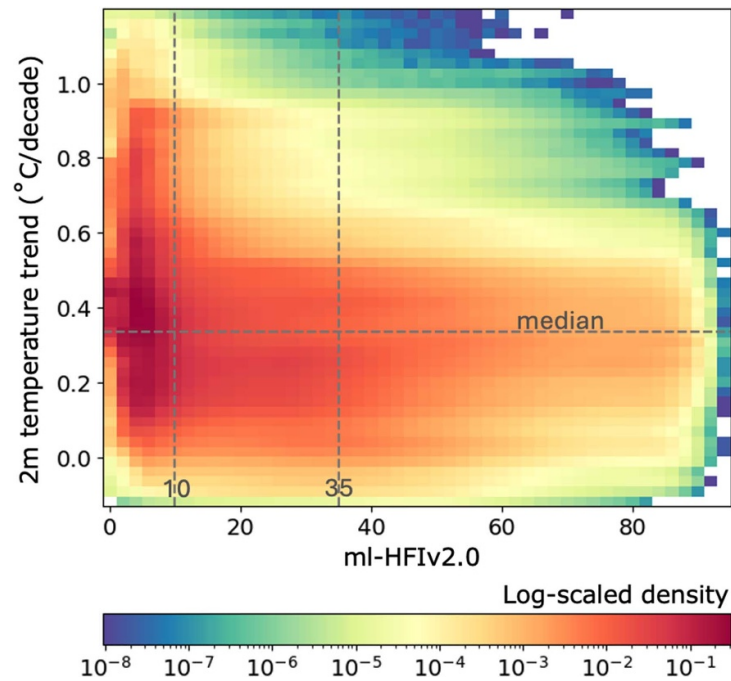


Figure 6. 2D density plot of 2024 ml-HFIv2.0 values for all land points globally versus their respective temperature trends from 1971–2023. Dashed vertical lines indicate thresholds for natural areas, used land, and settlements (see methods), while the dashed horizontal line represents the median 2 m temperature trend.

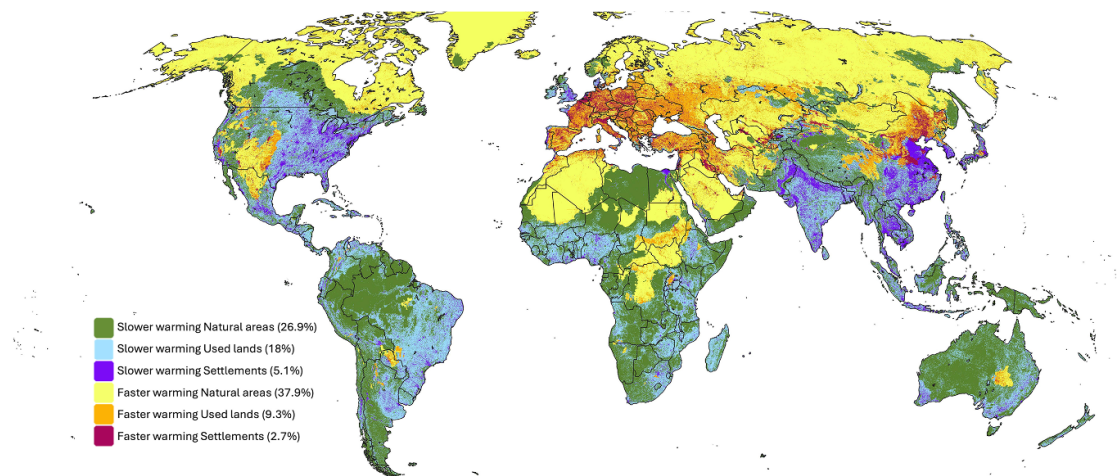


Figure 7. Global map showing the intersection of ml-HFIv2.0 classifications and temperature trends, categorizing land points by ml-HFIv2.0 thresholds (10 and 35) and the median temperature trend (dashed lines in figure 6).

While this study does not aim to investigate the drivers of these temperature trends and spatial distribution, it utilizes them to perform a combined analysis with human pressure on land, as quantified by the ml-HFI.

3.4. Combined analysis of human pressure and temperature trends

We categorize regions into six groups based on ml-HFIv2.0 thresholds (distinguishing natural areas, used lands, and settlements; see Methods) and the 50th percentile for above- and below-median warming rates. Figure 6 illustrates the distribution of pixels across categories while figure 7 helps to identify key spatial patterns of categories where human activities and climate impacts converge. The combined analysis of ml-HFIv2.0 and temperature trends provide a nuanced perspective on intersecting human-environment pressures (i.e. human-induced changes in climate and human-induced changes in the land surface). Here, we briefly describe these six categories.

- Slower warming natural areas (26.9%): Predominantly located in parts of Africa, South America, and sections of boreal forests, these regions indicate ecosystems with lower human pressure and slower rates of climate change. Oceanic and rainforest zones may naturally buffer temperature increases, also fall within this category. Their slower warming rates present critical opportunities for conservation, resilience-building, and biodiversity protection.
- Faster warming natural areas (37.9%): In contrast, a significant portion of natural areas are experiencing faster warming (primarily in the Arctic and northern forests) highlights ecosystems under threat despite low human pressure. These regions are undergoing rapid climate changes, underscoring the urgent need for targeted conservation and climate adaptation strategies.
- Slower warming used lands (18%): Used lands with slower warming trends, found in North America and Europe. These areas may reflect the influence of regional climate processes, certain land management, temperate climates, specific land characteristics, or other factors associated with lower warming rates, serving as reference points for understanding potential mitigation influences.
- Faster warming used lands (9.3%): Though a smaller category, identifying spatial patterns in these regions may help uncover correlations between land-use characteristics and higher warming rates.
- Slower warming settlements (5.1%): Most urban areas fall within the slower warming settlement category, predominantly located in eastern US, southern and southeastern Asia. Characterizing these areas alongside faster-warming settlements may shed light on patterns of urban development and climate interactions
- Faster warming settlements (2.7%): Urban settlements experiencing faster warming are relatively sparse, occurring in parts of East Asia and some European cities. These urban areas may be more susceptible to warming. Investigating the local conditions and urban characteristics, could help understand patterns.

When analyzing the percentage distribution of land categories across ml-HFIv2.0 and temperature change sections (section 3.4), some noteworthy information emerges. The most prominent finding is that natural areas are experiencing disproportionately faster warming, accounting for 37.9% of the total land area. Within the natural land category, this represents 58.5% of natural regions warming faster than the global median. In contrast, settlements exhibit the smallest proportion of rapid warming, accounting for only 2.7% of the total globe area while the remaining settlements, which are warming more slowly than the global median, account for an additional 5.1%. Within settlements, this translates to 35% experiencing rapid warming and 65% experiencing slower warming. Similarly, used lands have a larger proportion of areas experiencing slower warming (18%) compared to faster-warming areas (9.3%) of the total area, similar to the pattern observed in settlements. Although used lands have higher absolute values of warming compared to settlements, the ratio between rapid and slow warming (approximately 35% rapid vs. 65% slow) is similar for both categories. This suggests that areas categorized as ‘human-influenced lands’ exhibit a consistent distribution between rapid and slow warming. It is important to emphasize that in every category, the described relationships are only observed and co-occurring and this study does not attempt to establish causal links.

4. Discussions and conclusions

Mapping anthropogenic land transformations globally remains an emerging field (Watson *et al* 2023). This study presents several advancements by introducing the ml-HFI, an up-to-date, 300 m resolution dataset designed for continuous monitoring of land-use changes with significant socio-economic and environmental implications. This tool is particularly valuable for policymakers and scientists seeking to dynamically address human impacts and explore their relationships with other variables, such as public health (Skinner *et al* 2023). The ml-HFI’s ability to provide timely and continuous data, dependent only on the availability of Landsat imagery, could support the development of effective land-use policies aimed at sustaining global ecological health.

While this framework ensures continuous assessment of human pressure, it is important to note potential data inconsistencies inherent in the Landsat archive. For instance, the Landsat 7 scan line corrector failure in 2003 introduced data gaps (Markham *et al* 2005), which could affect the quality of predictions for certain regions and time periods. However, the design of the ml-HFIv2.0 approach allows predictions to be made using any available Landsat imagery, enabling seamless integration of newer Landsat data. This flexibility ensures that human pressure can be assessed both retrospectively and for the latest year, as long as imagery is available. Another limitation of the current version of ml-HFIv2.0 is its reduced predictive accuracy in regions with unique environmental or land-use characteristics. For example, the ml-HFI accuracy typically decreases in both rocky deserts and rocky mountainous areas

(e.g. figure S1(b)), perhaps due to a limited training sample size for these regions. While enhanced computational resources could possibly improve performance in these regions, the model already demonstrates robust accuracy for most landscapes.

This study uncovers critical patterns of combined human pressure via land use change and global warming, providing new insights into how human activities and climate change overlap across diverse landscapes. By integrating the ml-HFiv2.0 with long-term temperature trends, we identify regions where human-driven land transformations and climate changes converge, with the potential for amplified socio-environmental risks (IPCC 2019). Understanding these overlapping pressures is essential for designing effective, context-specific strategies to mitigate environmental degradation and enhance climate resilience (IPCC 2019).

By juxtaposing ml-HFiv2.0 with temperature trends, key regions where human influence and temperature change converge are identified. This emphasizes the need for region-specific strategies while revealing diverse challenges and opportunities across landscapes. For instance, regions with high human influence and accelerated warming, such as urban areas, emerge as priority targets for intervention (e.g. Gao *et al* 2024). Urban heat management through green infrastructure, emission reduction technologies, and sustainable urban planning could mitigate these dual pressures (e.g. Das *et al* 2024). Conversely, areas with lower human pressure demand urgent consideration for protecting vulnerable ecosystems from external pressures (Jacobson *et al* 2019) especially regions with low temperature trends that can serve as refugia from anthropogenic disturbances. We note that more densely-settled areas with lower temperature trends warrant investigation as this finding may suggest the presence of effective heat management strategies (e.g. Vargo *et al* 2016). In all cases, targeted strategies are necessary to balance development with climate resilience.

The existence of areas with higher development that are warming more slowly aligns with the findings of Sengupta *et al* (2024), who reported that regions with a low Human Development Index (HDI) experience more accelerated warming compared to high-HDI regions. Although their study used a different index and focused on 2015 data, their results remain consistent with ours. Additionally, they suggest that aerosols may be masking warming in human-influenced areas, implying that reductions in aerosol emissions could trigger abrupt temperature increases. Our findings offer valuable insights into the uneven distribution of warming across regions with different levels of human pressure, highlighting the need for further research into how human activities, land-use patterns, and atmospheric processes interact to shape regional climate trends, which remains a largely underexplored topic in the current literature.

The framework presented in this study is exploratory and descriptive, and it does not imply direct causal relationships between land pressure and temperature trends. However, the high-resolution of the ml-HFiv2.0 data enables (currently scarce) regional studies where detailed information can be used to draw specific conclusions about the relationship between human pressure on land and climate (e.g. Khelifa *et al* 2022, Sengupta *et al* 2024). This dataset also enables targeted regional studies that can inform policies to enhance ecological resilience, guide environmental planning, mitigate climate risks, and promote sustainable development.

Data availability statement

Landsat satellite data used in this study were obtained from the United States Geological Survey (USGS) and available through the Google Earth Engine platform at <https://developers.google.com/earth-engine/datasets>.

The ERA5-Land reanalysis dataset used in this study was obtained from the European Centre for Medium-Range Weather Forecasts (ECMWF) via the Copernicus Climate Data Store at <https://cds.climate.copernicus.eu/>.

All data and code used in this study are publicly available. The ml-HFiv2 dataset can be explored and accessed through the ml-HFiv2 web application (<https://mlhfi.com>) and the data only is hosted on Dryad (DOI:<https://doi.org/10.5061/dryad.m63xsj4fk>). The Google Earth Engine scripts used for Landsat preprocessing (DOI:<https://doi.org/10.5281/zenodo.17452372>) and the machine learning model code (DOI:<https://doi.org/10.5281/zenodo.17670299>) are available.

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ORCID iDs

Bryam Orihuela-Pinto  0000-0003-3873-9401

Patrick W Keys  0000-0002-7250-1563

Frances V Davenport  0000-0002-3061-2062

Elizabeth A Barnes  0000-0003-4284-9320

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